

Using Machine Learning to Predict Antimicrobial Resistance (AMR) Phenotypes

Namkyeong Kim, Yeojin Jung, Donghoon Kim
Arkansas State University, USA.

Motivation

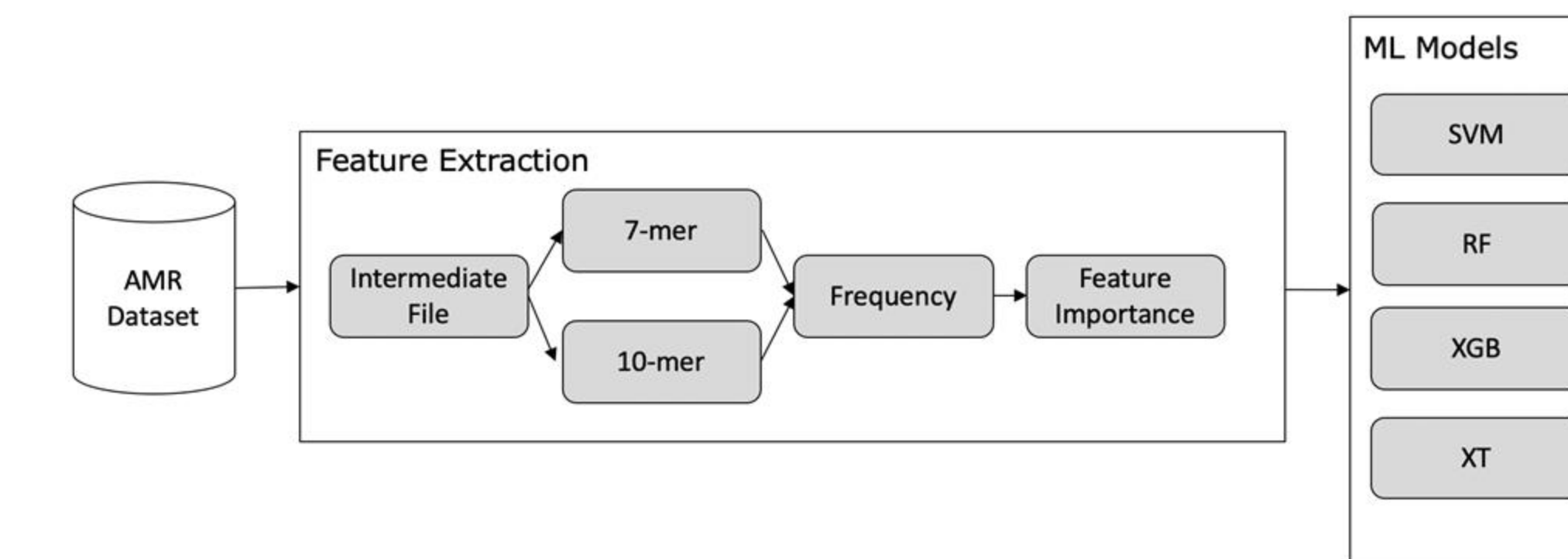
- Antimicrobial resistance (AMR) is a critical global public health concern due to the rise in hospital acquired infections and heightened levels of illness and death.
- Researchers can utilize ML techniques to predict AMR phenotypes (resistant or susceptible)
- The objective of this study is to enhance AMR prediction using ML methods utilized in the cybersecurity domain due to the similarities between AMR and malware datasets.

Dataset & Features

- AMR metadata collection by ~BV-BRC (Bacterial and Viral Bioinformatics Center)
 - In genome name column, *Enterococcus faecium*
 - In antibiotic column, vancomycin
 - classify “resistant” and “susceptible”
- Feature Design
 - 562.48345.fna -> 562.48345.fna.txt

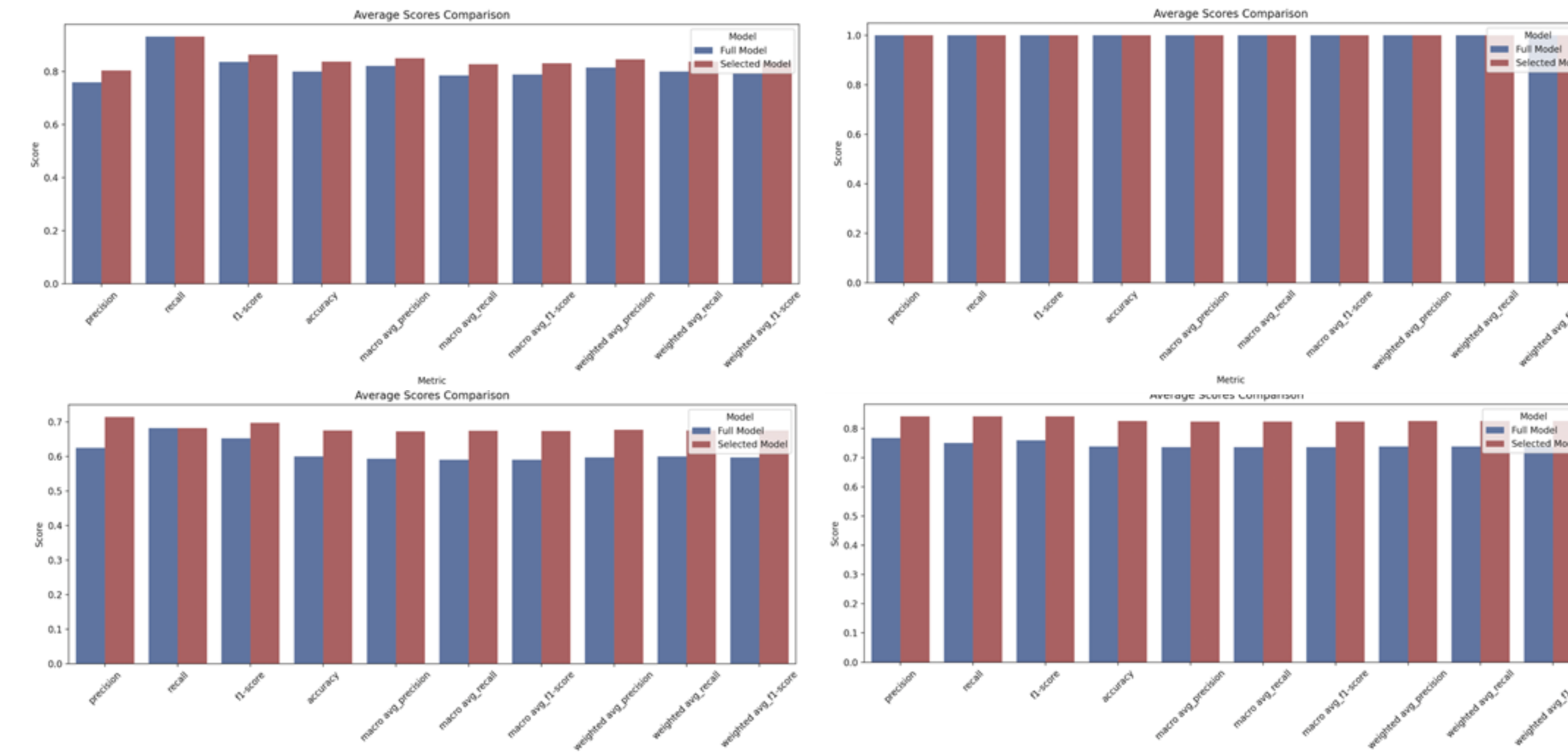
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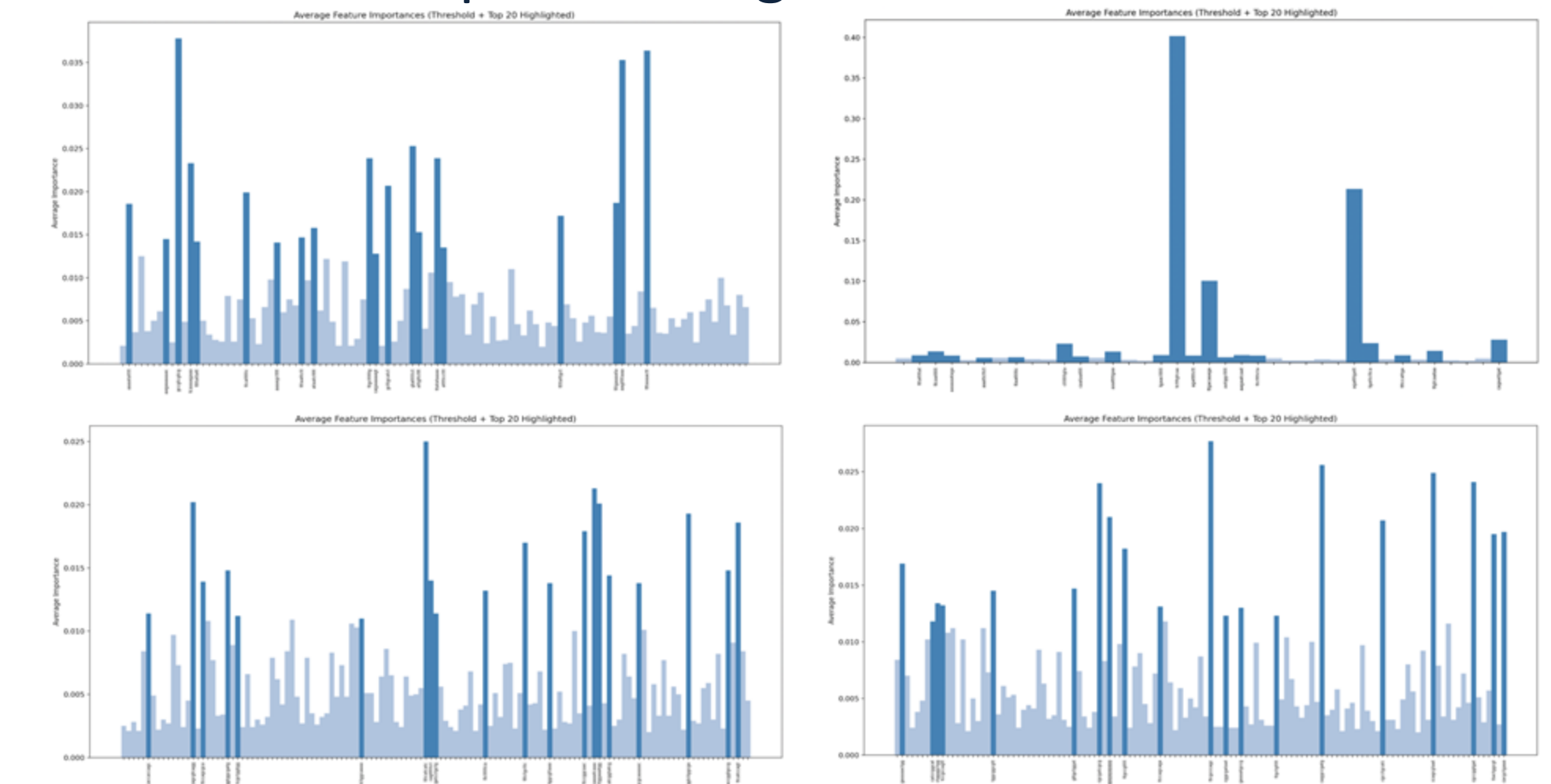


- We utilize n-gram frequency analysis used in malware detection
 - Both dataset have similar characteristics
 - Limited number of opcode vs. 4 DNA bases (A, T, G, C)
- The detailed step
 - ☐ Step 1. Download AMR Dataset files from the BV-BRC repository for resistant and susceptible cases.
 - ☐ Step 2. Select the number of files each from the resistant and susceptible cases in order of file size out of over 2,000 files.
 - ☐ Step 3. Make intermediate files by filtering out meta information.
 - ☐ Step 4. Generate k-mers (i.e., 7 or 10-mers) from each file.
 - ☐ Step 5. Calculate the frequency of each k-mers across all files and select the top number features out of 1 million features.
 - ☐ Step 6. Generate CSV files containing the top number features with labels assigned as 0 for resistant and 1 for susceptible.
 - ☐ Step7. Use ensemble ML models to analyze the data

- Comparison of accuracy between using all features and using 1,000 features selected based on feature importance



- Feature Importance for Top 1,000 frequency exceeding a set threshold and Top 20 among them



- High frequency features may not be an high feature importance

Analysis

- Table 1 presents the experimental results using 4 datasets, highlighting the best-performing model and corresponding F1-scores for each dataset

Table 1: Experimental results using 4 datasets

Genome Name	Antibiotic	Resistant	Susceptible	Dataset	Feature	Model	F1-score
<i>E. faecium</i>	vancomycin (VAN)	1,229	1,146	200	1,000	XGB	92.31%
<i>E. faecium</i>	ampicillin (AMP)	531	209	200	1,000	RF/XGB	100.00%
<i>E. coli</i>	trimethoprim/sulfamethoxazole (SXT)	1,081	2,477	200	1,000	XGB	69.77%
<i>E. coli</i>	ampicillin (AMP)	3,026	2,577	200	1,000	XGB	84.09%

- ❑ Analysis of experimental results
 - ❑ The results indicate that XGB is the optimal model across all 4 datasets.
 - ✓ *E. faecium* with VAN with F1-score of 92.31%
 - ✓ *E. faecium* with AMP with F1-score of 100.00%
 - ✓ *E. coli* with SXT with F1-score of 69.77%
 - ✓ *E. coli* with AMP with F1-score of 84.09%

Conclusion

- This research has demonstrated that the performance of AMR prediction can be enhanced by applying cross domain research methodologies that leverage the similar characteristics found in different datasets.
- TThis study has a limitation due to the use of a small dataset for the experiments. We plan to continue by utilizing a larger dataset moving forward.